# A Semi-Supervised Semantic and Spatial Change Detail Retention Network for Semantic Change Detection in Remote Sensing Images

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***Abstract*— The task of semantic change detection (SCD) in remote sensing images (RSIs) is aimed at identifying the multiple "from–to" changes in land cover. However, the prevailing multitask SCD network structure does not directly model the semantic change information, leading to potential inaccuracies in extracting multiple detailed changes. In this article, a semi-supervised semantic and spatial change detail retention network ($S^3$CDRNet) is proposed to precisely model the details of spatial changes and the distribution of multiple change categories for RSIs. The encoder of the proposed $S^3$CDRNet is based on a Siamese structure, where a shallow convolutional neural network (CNN) followed by a transformer are used to extract the local–global change features. A precise semantic change perception module (PSCPM) based on large kernel convolution is then introduced to enhance the weak semantic changes. The decoder consists of several deconvolution layers to restore the original resolution, and a pixelwise semantic change map is acquired. To further distinguish the inherent imbalanced change types, an adaptive category-balanced semi-supervised learning (ACBSS) strategy is developed to better use the abundant class distribution information from the unlabeled image pairs. In the experiments conducted in this study, we made an in-depth study of multiple SCD based on three public datasets: the SECOND dataset, the Landsat-SCD dataset, and the Hi-UCD mini dataset. The results show the potential of the proposed method under scenarios with various change classes and an imbalanced change class problem, compared with the state-of-the-art SCD methods.**

***Index Terms*— Deep neural network, remote sensing images (RSIs), semantic change detection (SCD), semi-supervised learning.**

## I. Introduction

THE change detection (CD) technology of multitemporal optical remote sensing images (RSIs) reflects the land surface change from a fine scale, and it is an important means to realize the automatic and intelligent monitoring of the land surface [1], [2]. Many different methods have been proposed in the field of binary CD (BCD), including supervised methods [3], [4], methods based on point supervision [5], methods based on vectorization of CD [6], semi-supervised methods [7], weakly supervised methods [8], and unsupervised methods [9], [10]. Compared with BCD, semantic CD (SCD) can provide the category transition information of the observed region and has become an important research field [11], [12], [13].

Traditional SCD methods are mainly based on combined feature extraction and machine learning algorithms, and the semantic changes are usually acquired by direct classification or postclassification comparison, in methods such as change vector analysis (CVA) [14], support vector machine (SVM) [15], decision tree classifier [16], object-based methods [17], and Bayes classifier [18].

Deep learning techniques have been the subject of much attention in SCD for their advantage in feature learning with big data. Some researchers have modeled SCD as a multitask learning problem, which is aimed at acquiring the bitemporal semantic segmentation results for the changed areas [19], [20], [21]. Caye Daudt et al. [19] proposed a triple-branch SCD network based on a convolutional neural network (CNN), which contains two semantic segmentation branches and one BCD branch, where the difference features from the two semantic segmentation encoder stages are extracted and then connected to the decoder with the BCD branch. Yang et al. [20] proposed a Siamese asymmetric network combining a series of asymmetric pyramid modules to consider the varied spatial characteristics of changes. Ding et al. [21] proposed a triple-branch network—Bi-SRNet—which introduces cosine similarity in the loss function to constrain the semantic consistency of the bitemporal segmentation results for the changed regions. Subsequently, the cosine similarity loss was improved for the unchanged regions in SCanNet [22]. Zheng et al. [23] proposed a multitask hybrid CNN–transformer architecture (ChangeMask), which is combined with a symmetric change feature fusion mechanism to preserve the discriminative information of change regions. In [24], the pseudosegmentation labels generated from one time phase were used to supervise the output semantic segmentation result of the other time phase to enhance the consistency of the semantics in unchanged areas. Deng et al. [25] introduced a contrastive loss function for a triple-branch CNN, which is based on prior knowledge that encourages the homogeneity of the unchanged regions and the heterogeneity of the changed regions. Xia et al. [26] changed BCD loss into SCD loss in the CD branch to further

Received 3 July 2024; revised 19 October 2024; accepted 4 November 2024. Date of publication 14 November 2024; date of current version 26 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42001307 and in part by the Natural Science Foundation of Ningxia under Grant 2024AAC05029. *(Corresponding author: Pengyuan Lv.)*

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Digital Object Identifier 10.1109/TGRS.2024.3497983

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consider the semantic change information in the network. Wang et al. [27] proposed a difference enhancement module for the SCD network to improve the extracted change information under different scales. In [28], a feature fusion module based on a spatial attention mechanism was introduced to better distinguish the foreground and background information in complex changing scenes. Tian et al. [29] proposed to use a spatial-channel attention mechanism to enhance the salient regions of a single-temporal RSI, which are prone to be the candidate changed regions. Cui and Jiang [30] used a Siamese Swin transformer encoder to extract multiscale features and introduced a two-level difference feature fusion strategy to acquire the multiple change information. Li et al. [31] proposed to improve the interaction of different subtasks in the decoder. Chang et al. [32] modified the existing triple-branch structure. The encoded bitemporal features were concatenated and then sent to the triple-branch decoder, so as to strengthen the correlation of each branch and perceive the bitemporal semantics. Another approach for multitask SCD is based on a dual-branch network structure. In [33], an SCD network based on Siamese U-Net was proposed, which combines multiscale difference features to preserve the detailed changes. Wen et al. [34] designed a dual-branch network based on 3-D ResUnet to perform the BCD and SCD tasks simultaneously for bitemporal RSIs. In parallel, a 3-D generative adversarial network (GAN) was also incorporated into the BCD branch to enhance the discriminative ability for potential high-level change features.

In contrast to the multitask approach, the end-to-end SCD approach directly models the mapping between the input bitemporal images and the semantic change map. Zhu et al. [35] constructed a Siamese architecture with shared weights to extract bitemporal image features separately. To avoid the influence of unchanged regions, the BCD results are added as change mask constraints at the decoder side. Yuan et al. [36] proposed a dual-branch pyramid transformer network to enhance the semantic changes. Wang et al. [37] constructed a dual-branch network based on a CNN and a graph neural network, which introduces a graph attention network combined with spatial location information to enhance the model's ability to understand more specific geospatial structures.

Although the existing SCD methods can acquire good results, there are still two challenging problems to be solved in the SCD task.

1) The first problem is the imbalance of the spatial characteristics of changed regions. As shown in Fig. 1(a), there are two change categories. One category is "low vegetation" to "ground," and the other is "low vegetation" to "building." Due to the difference in the size and shape of the semantics in each single-temporal image, there is a more significant difference in the spatial characteristics of the changed regions in bitemporal images, which brings more challenges in predicting the precise spatial change details.
2) The second problem is the imbalance of the change types. In Fig. 1(b), the number of pixels for the 17 change classes in the SECOND dataset is displayed. It can be seen that there is a strong long-tail problem, which indicates that it is difficult to acquire enough samples for the tail categories in the training phase, and the ability of the network to distinguish more change categories is limited.

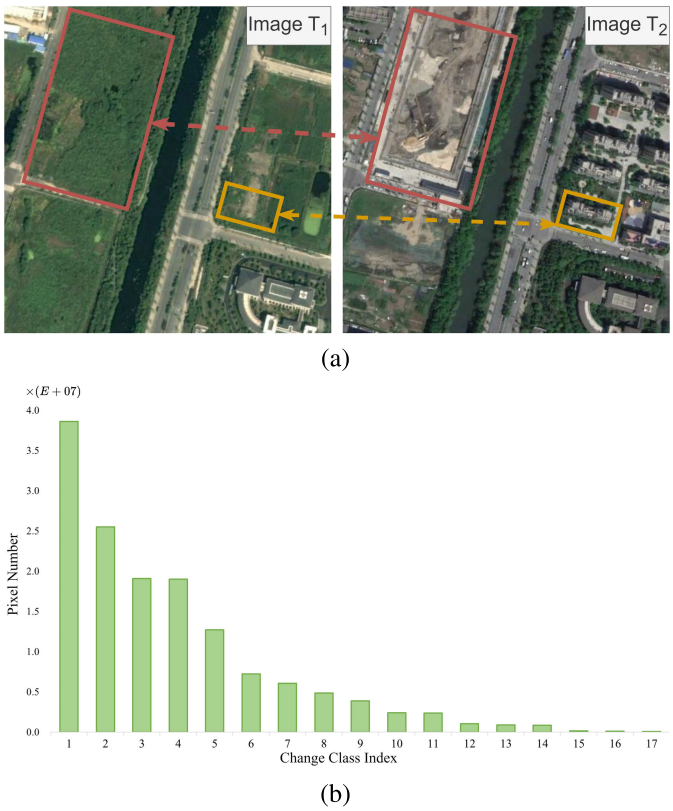



Fig. 1. Spatial size imbalance and class imbalance problem in the SECOND dataset. (a) Spatial size imbalance in changed regions. (b) SECOND dataset class imbalance problem.

Considering the above problems, we chose an end-to-end neural network structure as the base model, which can express the semantic changes more directly. The major contributions of this work are as follows.

1) We propose a spatial change detail retention SCD network (SCDRNet) to preserve the detailed spatial change information at different scales. The Siamese CNN–transformer hybrid structure encoder is designed to express the local–global change information. A precise semantic change perception module (PSCPM) is also proposed, based on large kernel convolution, to effectively introduce the background at a larger range, which is beneficial in predicting small change targets.
2) An adaptive category-balanced semi-supervised learning (ACBSS) strategy is developed for SCDRNet to better use the abundant distribution information in the unlabeled image pairs. The ACBSS strategy is based on the mean teacher framework, and an adaptive class sampling mechanism is introduced to consider the class imbalance problem.
3) In the experimental section, a further in-depth study of the ability of the SCD network in modeling various change classes and handling the imbalanced change

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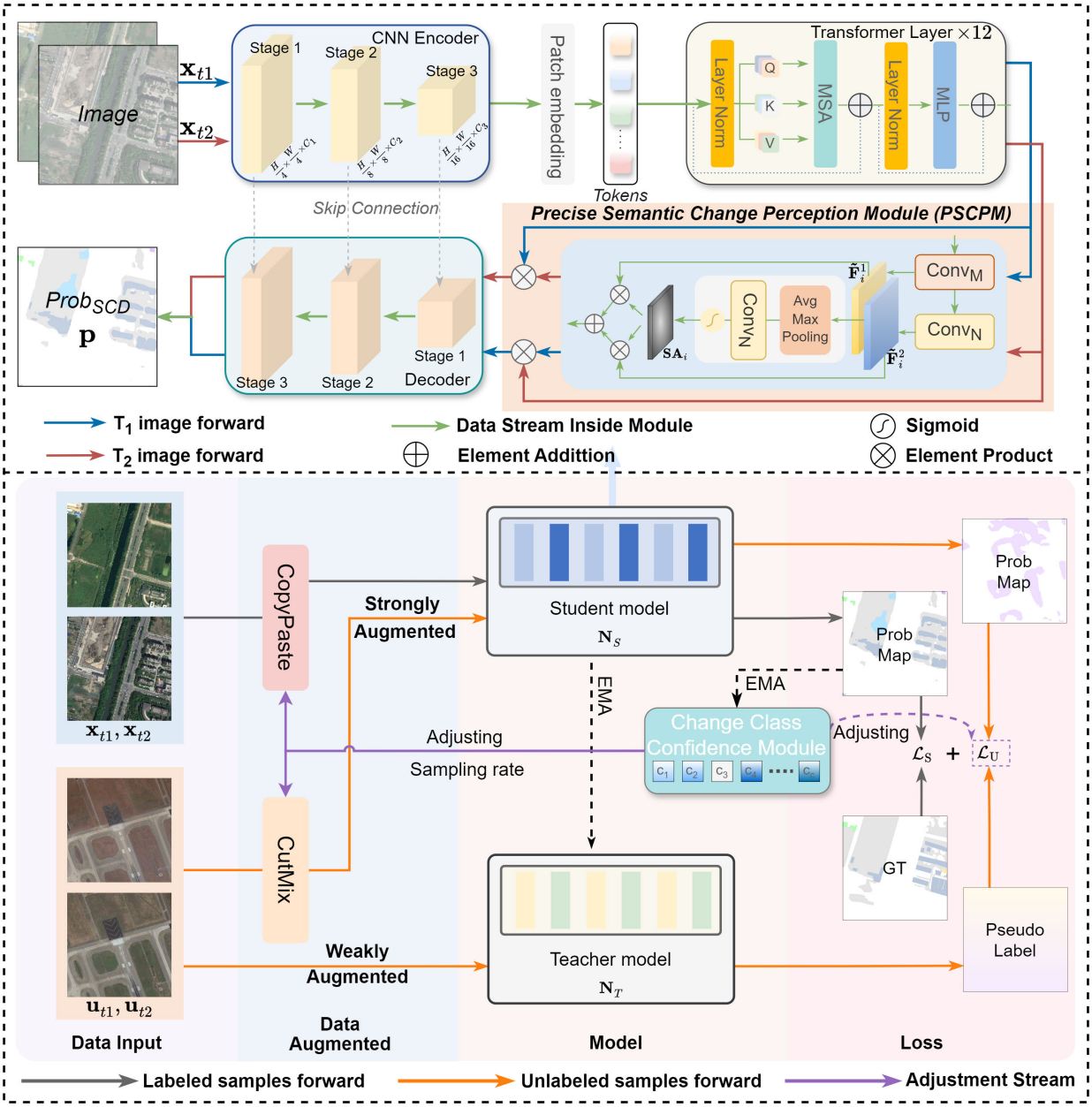



Fig. 2. Architecture of S$^3$CDRNet.

class problem has been provided. The existing multitemporal land-use/land-cover labels in the public SCD datasets were further extended into multiple semantic change labels (up to 18 change categories) through label differencing and mapping, providing the data basis for analyzing the above-mentioned problems.

The rest of this article is organized as follows. In Section II, we introduce the proposed method. Section III introduces the experimental settings, and an analysis is provided in Section IV. Finally, the conclusion of this study is provided in Section V.

## II. Proposed Method

The overall flowchart of the proposed S$^3$CDRNet is shown in Fig. 2. First, the network structure is introduced. The proposed S$^3$CDRNet is based on a Siamese structure. In the encoder, a shallow CNN is used to extract preliminary semantic features, and a vision transformer [38] is used to further extract the global deep expression of change features. The PSCPM based on large kernel convolution and spatial attention is then introduced to preserve the fine-scale semantic changes. The large kernel convolution increases the model receptive field and encourages the model to perceive the background semantic information of small target regions [39]. For the decoder, a symmetrical U-shaped deconvolution structure is designed to restore the original resolution of the features, and skip connections are added to protect the multiscale local details of the bitemporal images. Finally, a soft-max function is used to acquire the pixelwise semantic change map. The second part is the ACBSS strategy. The ACBSS strategy uses the pseudolabel-based mean teacher model as the

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basic framework. The labeled samples are sent to the student model after data augmentation (Copy–Paste [40]), and the supervised loss function is constructed based on cross-entropy loss. The unlabeled samples are sent to the teacher model to generate pseudolabels after weak augmentation and are then used to supervise the output of the student model after strong augmentation (CutMix [41]). Furthermore, we introduce a change class confidence module to assess the accuracy of the prediction results through the student model. Based on this, we determine the confidence of each class and adjust the sampling rate of the two data augmentation strategies accordingly. This encourages the increase in the sampling rate for the change types with low detection accuracy in the augmentation step to provide more samples for the hard classes in the training phase.

### A. Spatial Change Detail Retention SCD Network (SCDRNet)

We let the bitemporal images $\mathbf{x}_{t1}, \mathbf{x}_{t2} \in \mathbb{R}^{3\times H\times W}$ be the input of the network. The encoder follows the Siamese CNN–transformer structure. The CNN block is adopted to acquire the shallow feature $\mathbf{F}_{ti}^{c} \in \mathbb{R}^{C\times H/16\times W/16}$, $i \in [1, 2]$. To further extract the deep semantic features of bitemporal RSIs, the shallow features $\mathbf{F}_{ti}^{c}$ are input into a 12-layer transformer [42], and the extracted bitemporal features are defined as $\mathbf{F}_{ti} \in \mathbb{R}^{512\times H/16\times W/16}$. Here, the calculation of each self-attention layer can be expressed as follows:

$$\hat{\mathbf{x}}^{l} = \mathrm{MSA}\left[\mathrm{LN}(\mathbf{x}^{l-1})\right] + \mathbf{x}^{l-1} \tag{1}$$

$$\mathbf{x}^{l} = \mathrm{MLP}\left[\mathrm{LN}(\hat{\mathbf{x}})^{l}\right] + \hat{\mathbf{x}}^{l} \tag{2}$$

where MSA stands for the multihead self-attention mechanism, LN stands for the layer normalization operation, MLP stands for the multilayer perceptron, and $\mathbf{x}^{l}$ represents the output token of the $l$th layer.

In RSIs, the spatial sizes of the changed regions vary a lot, and some targets can be relatively small. Although small changed features extracted by the model can be easily disturbed by the semantic changes in the surrounding objects, it is helpful to use the correspondence information with the background to distinguish small changed targets. This prior knowledge can improve the accuracy of the model on the semantic features of small changed regions. Considering this prior knowledge, the proposed PSCPM uses large kernel convolution to increase the spatial receptive field, to better perceive the spatial context information of various backgrounds around small changed objects.

The PSCPM is also based on the Siamese structure. We let $\mathbf{F}_{ti}$ be the input to the PSCPM, $\hat{\mathbf{F}}_{i}^{1}$ and $\hat{\mathbf{F}}_{i}^{2}$ are the output after two cascaded large kernel convolution operations, and $\hat{\mathbf{F}}_{i}$ is the final output after convolution and concatenation operations

$$\hat{\mathbf{F}}_{i}^{1} = \mathrm{Conv}_{M}(\mathbf{F}_{ti}) \tag{3}$$

$$\hat{\mathbf{F}}_{i}^{2} = \mathrm{Conv}_{N}\left(\hat{\mathbf{F}}_{i}^{1}\right) \tag{4}$$

$$\hat{\mathbf{F}}_{i} = \mathrm{Concat}\left(\hat{\mathbf{F}}_{i}^{1}, \hat{\mathbf{F}}_{i}^{2}\right) \tag{5}$$

where $\mathrm{Conv}_{M}(\bullet)$ and $\mathrm{Conv}_{N}(\bullet)$ represent the convolution operation with the kernel size of $M \times M$ and $N \times N$, respectively. Concat$(\bullet)$ is the concatenation operation.

A spatial attention mechanism then follows to better focus on the object change features. First, the features are extracted through pooling

$$\mathbf{SA}_{i}^{\mathrm{avg}} = \mathrm{AvgPool}(\hat{\mathbf{F}}_{i}) \tag{6}$$

$$\mathbf{SA}_{i}^{\mathrm{max}} = \mathrm{MaxPool}(\hat{\mathbf{F}}_{i}) \tag{7}$$

where AvgPool$(\bullet)$ is the average pooling, and MaxPool$(\bullet)$ represents the maximum pooling. To make different spatial features interact with each other, the two sets of spatial pooling features are concatenated, and convolution is used to convert the pooling features into a spatial attention weight

$$\mathbf{SA}_{i} = \mathrm{Sigmoid}\left\{\mathrm{Conv}_{N}\left[\mathrm{Concat}\left(\mathbf{SA}_{i}^{\mathrm{avg}}, \mathbf{SA}_{i}^{\mathrm{max}}\right)\right]\right\}. \tag{8}$$

The feature map from the large kernel convolution sequence is weighted by its corresponding spatial attention weight $\mathbf{SA}_{i}$, and the attention feature $\mathbf{S}_{i}$ is obtained as

$$\mathbf{S}_{i} = \mathrm{Conv}\left(\mathbf{SA}_{i} \cdot \hat{\mathbf{F}}_{i}^{1} + \mathbf{SA}_{i} \cdot \hat{\mathbf{F}}_{i}^{2}\right). \tag{9}$$

Finally, the concept of cross-attention [43] is used to model the temporal correlation, thereby enhancing the cross-temporal semantic changes. The output feature $\mathbf{F}_{i}^{\mathrm{PSCPM}}$ is acquired based on (10):

$$\mathbf{F}_{k}^{\mathrm{PSCPM}} = \mathbf{S}_{k} \cdot \mathbf{F}_{ti} \tag{10}$$

where $i = 1, 2$, $k = 2, 1$. Through large kernel convolution modules, the perception ability of SCDRNet with subtle changes can be improved.

At the decoder, three upsampling stages using deconvolution are used to restore the change features to their original resolution. Feature skip connections are performed to enhance the spatial details of the features.

### B. Adaptive Category-Balance Semi-Supervised Learning (ACBSS) Strategy

The ACBSS strategy is based on the mean teacher framework, where the labeled data are defined as $\mathcal{D}^{l} = \{(\mathbf{x}_{t1}^{i}, \mathbf{x}_{t2}^{i}, \mathbf{y}^{i})\}_{i=1}^{M}$. $M$ represents the number of samples in the labeled set. In the same way, the unlabeled data are defined as $\mathcal{D}^{u} = \{(\mathbf{u}_{t1}^{j}, \mathbf{u}_{t2}^{j})\}_{j=1}^{N}$, where $N$ represents the number of samples in the unlabeled set. Specially, we define the student SCDRNet model as $\mathrm{N}_{S}$, and $\mathrm{N}_{T}$ represents the teacher SCDRNet model.

First, for the labeled samples, data augmentation (Copy–Paste) is applied to the image pair $(\mathbf{x}_{t1}, \mathbf{x}_{t2})$, the enhanced image $(\mathbf{x}_{t1,\mathrm{Aug}}, \mathbf{x}_{t2,\mathrm{Aug}})$ is input to the student model $\mathrm{N}_{S}$, and the change probability map $\mathbf{p}$ is acquired

$$\mathbf{p} = \mathrm{N}_{S}(\mathbf{x}_{t1,\mathrm{AUG}}, \mathbf{x}_{t2,\mathrm{AUG}}). \tag{11}$$

The supervised loss $\mathcal{L}_{S}$ is then computed based on the following equation:

$$\mathcal{L}_{s} = \frac{1}{N_{l}} \sum_{k=1}^{N_{l}} \frac{1}{WH} \sum_{j=1}^{WH} \ell_{\mathrm{ce}}(\mathbf{p}_{kj}, \mathbf{y}_{kj}) \tag{12}$$

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where $N_l$ represents the number of labeled samples in the training batch, $W$ and $H$, respectively, represent the width and height of the input samples, $\ell_{\text{ce}}(\bullet)$ denotes the pixelwise cross-entropy loss function, $\mathbf{p}_{kj}$ is the prediction of the $j$th pixel in the $k$th sample, and $\mathbf{y}_{kj}$ represents the corresponding true label, using one-hot encoding.

Then, for unlabeled samples, both strong and weak data augmentation (CutMix) are applied to the unlabeled image pair $(\mathbf{u}_{t1}, \mathbf{u}_{t2})$. The image-level perturbations help the network to better learn the intrinsic spectral and spatial distributions of the imagery [44]. Furthermore, it can help mitigate the class imbalance problem by generating more tail-class samples. The weakly augmented data $(\mathbf{u}_{t1,w}, \mathbf{u}_{t2,w})$ are input into the teacher model $\mathrm{N}_T$ to generate the pseudolabel $\mathbf{y}_p$. The strongly augmented data $(\mathbf{u}_{t1,s}, \mathbf{u}_{t2,s})$ are input into the student model $\mathrm{N}_S$ to generate the semantic change prediction rules $\mathbf{p}_s$, and the dynamic consistency loss $\mathcal{L}_U$ is constructed from $\mathbf{y}_p$ and $\mathbf{p}_s$

$$\mathcal{L}_u = \frac{1}{N_u}\sum_{k=1}^{N_u}\frac{1}{\sum_{j=1}^{WH} w_{kj}}\sum_{j=1}^{WH} w_{kj}\ell_{\text{ce}}(\mathbf{p}_{kj}, \mathbf{y}_{Pkj}) \tag{13}$$

$$w_{kj} = \max_{c\in\{1,\ldots,C\}} \left(\mathbf{p}_{kj}^c\right)^{\gamma}\mathbb{Z}_{kj} \tag{14}$$

where $N_u$ represents the number of unlabeled samples in the training batch, and $w$ is introduced to adaptively assign more weight to the samples with reliable predictions for the augmented patches. $\mathbb{Z}_{kj} \in \{0, 1\}$ indicate whether the pixel is from the augmented patch or not. $\gamma$ is the adjustable parameter.

The total loss $\mathcal{L}_{\text{total}}$ consists of the supervised loss $\mathcal{L}_S$ and the consistency loss $\mathcal{L}_U$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_S + \mathcal{L}_U. \tag{15}$$

To address the class imbalance problem, the ACBSS strategy introduces a change class confidence module to record the prediction confidence of each class for the labeled samples during each training iteration [45]. First, the class confidence index is defined as follows:

$$\alpha^c = \frac{1}{N_l}\sum_{k=1}^{N_l}\frac{1}{N_k^c}\sum_{j=1}^{N_k^c}\mathbf{p}_{kj}^c, \quad c \in \{1, \ldots, C\} \tag{16}$$

where $\mathbf{p}_{kj}^c$ denotes the prediction of the $c$th class of the $j$th pixel in the $k$th sample. $N_k^c$ denotes the number of pixels belonging to class $c$. The exponential moving average (EMA) algorithm is then used to update the value of $\alpha^c$ for each training iteration

$$\alpha_{\text{Iter}}^c \leftarrow \tau\alpha_{\text{Iter}-1}^c + (1-\tau)\alpha_{\text{Iter}}^c, \quad c \in \{1, \ldots, C\} \tag{17}$$

where Iter denotes the Iterth iteration, and $\tau$ is the momentum coefficient, which is set to 0.999.

The class confidence index $\alpha^c$ is then used to adjust the sampling rate $r \in \mathbb{R}^C$ in the data augmentation step

$$r^c = \frac{e^{(1-\alpha^c)}}{\sum_{j=1}^{C} e^{(1-\alpha^j)}}, \quad c \in \{1, \ldots, C\}. \tag{18}$$

Equation (18) indicates that if the confidence $\alpha$ of a particular class is low, indicating unreliable predictions, the sampling rate $r^c$ for this class should be increased. Conversely, if the confidence is high, the sampling rate should be decreased.

## III. Experimental Settings

### A. Descriptions of Datasets

*1) SCD Dataset (SECOND) [20]:* The SECOND dataset presents a significant challenge for SCD. The dataset is made up of 4662 pairs of high-resolution optical images with resolutions ranging from 0.5 to 3 m, captured over the cities of Chengdu, Shanghai, and Hangzhou in China, by various sensors. In total, 2968 image pairs of the whole dataset are accessible and were used in the experiments. The spatial size of each image in the SECOND dataset is 512 × 512 pixels, and there are six semantic categories. For the change categories, there are, theoretically, 31 possible combinations. After excluding the nonexistent and extremely rare categories, 18 change categories remained for the experiments. The dataset was randomly partitioned into training and test sets, with the training set containing 2375 image pairs and the test set containing 593 image pairs.

*2) Landsat-SCD Dataset [36]:* The Landsat-SCD dataset comprises 2385 original pairs of 30-m resolution Landsat images captured over the city of Tumushuke in China from 1990 to 2020. Each image pair is 416 × 416 pixels in size. The Landsat-SCD dataset contains four semantic categories, theoretically resulting in 13 possible change combinations. After excluding the nonexistent categories, ten change categories were used for the experiments. The dataset was randomly partitioned into training and test sets, with the training set containing 1908 image pairs and the test set containing 477 image pairs.

*3) Hi-UCD Mini Dataset [46]:* The Hi-UCD mini dataset comprises 1293 pairs of 0.1-m resolution optical images captured in Tallinn, Estonia, between 2017 and 2019. The size of each image is 1024 × 1024 pixels. This dataset contains nine semantic categories, theoretically allowing for 73 possible combinations. Due to the insufficient number of samples for some change classes to support the training of the model, eight types of change classes were retained. Each original image was cropped into four nonoverlapping smaller images of 512 × 512 pixels. After discarding invalid samples, a total of 882 images were obtained. These images were then randomly divided into a training set containing 794 image pairs and a test set with 88 image pairs.

### B. Implementation Details

The hyperparameter settings and the training details of the comparison methods used in the experiments are provided in this part.

*1) Network Structure and Parameters of S$^3$CDRNet:* In the CNN–transformer hybrid encoder, the CNN encoder uses the first three stages of ResNet-50 to extract shallow features. Following the CNN structure, a 12-layer transformer is integrated to capture global features. The kernel sizes of $\text{Conv}_M(\bullet)$ and $\text{Conv}_N(\bullet)$ in PSCPM were set to 7 and 9, respectively, based on the experimental analysis. In the decoder, we designed a three-stage deconvolution decoder with a symmetrical U-shaped structure to restore the original resolution of the features. Meanwhile, skip connections are incorporated to

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preserve multiscale details. In the ACBSS strategy, the pseudolabel threshold $\tau_p$ is a crucial parameter that determines the quality of the generated pseudolabels. In this study, parameter $\tau_p$ was set to 0.7, based on the experimental analysis.

*2) Training Settings:* The experiments described in this article were conducted on a workstation equipped with two parallel NVIDIA GeForce 3090 GPUs and implemented in the Python environment. For the semi-supervised methods, the same parameters were used for the three datasets, where the batch size was set to 2, the training epochs were set to 200, and the initial learning rate was set to 0.01. Stochastic gradient descent (SGD) with Nesterov momentum was used as the optimizer, with a weight decay of 0.0001. Random flipping and rotation were used as the weak augmentation in the semi-supervised learning and fully supervised learning.

### C. Evaluation Metrics

The mean intersection over union (mIoU) and mean $F1$-score are used here as indicators to evaluate the network performance.

The mIoU calculates the intersection ratio between the predicted result and the true mask

$$\text{mIoU} = \frac{1}{c+1}\sum_{i=1}^{c}\frac{\text{TP}}{\text{FP}+\text{FN}+\text{TP}} \tag{19}$$

where TP, FP, and FN are the numbers of true positives, false positives, and false negatives, and $c$ represents the number of the change classes.

The $F1$-score is the harmonic average of the precision and recall, which can evaluate the accuracy of prediction results in a more balanced way

$$\text{precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{20}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{21}$$

$$F1\text{-score} = \frac{2\times\text{precision}\times\text{recall}}{\text{precision}+\text{recall}}. \tag{22}$$

## IV. Experimental Results and Analysis

### A. Ablation Study and Parameter Analysis

*1) Ablation Analysis of the Proposed Modules:* The effectiveness of the PSCPM and ACBSS modules was analyzed using the SECOND dataset, the Landsat-SCD dataset, and the Hi-UCD mini dataset, with 20% of training samples.

The accuracy results are presented in Table I. For the SECOND dataset, the first two rows in the table illustrate the supervised conditions. Notably, the inclusion of the PSCPM results in a 0.71% increase in mIoU and a 0.84% improvement in $F1$-score. The improvement in mIoU indicates that the PSCPM plays a significant role in enhancing the model's ability to perceive subtle semantic changes. Using large kernel convolution, the PSCPM effectively expands the receptive field of the model, incorporating a broader range of background contextual information, which enhances the ability to distinguish different changed objects. The third and fourth rows represent the semi-supervised conditions. In the semi-supervised conditions, the addition of the PSCPM results in a 1.91% increase in mIoU and a 2.96% improvement in $F1$-score. The improvement of adding the PSCPM in the semi-supervised phase is larger than in the supervised phase. This can be attributed to the fact that the ACBSS strategy is capable of leveraging a larger number of unlabeled samples to train the PSCPM, enabling it to better capture the features of subtle semantic change. Consequently, the model's generalization ability is enhanced. The above results demonstrate the significant impact of the PSCPM in enhancing the model performance in detail preservation and denoising under both supervised and semi-supervised learning conditions. Comparing the second and fourth rows, the semi-supervised strategy of the ACBSS strategy increases the mIoU and $F1$-score by 1.99% and 2.66%, respectively. For the Landsat-SCD dataset, when comparing the first row and the second row, the introduction of the PSCPM leads to a significant improvement of 5.79% in mIoU and 5.59% in $F1$-score. This improvement is notable because the Landsat-SCD dataset has a lower resolution, resulting in smaller scales for some objects, such as "river depletion" and "newly constructed buildings," and the proposed PSCPM is effective on these small changes. Comparing the second row with the fourth row, the incorporation of the ACBSS strategy allows the model to optimize its parameters using unlabeled samples, leading to an additional increase of 1.91% in mIoU and 1.65% in $F1$-score. For the Hi-UCD mini dataset, the performance is generally similar to the performance with the SECOND and Landsat-SCD datasets. With the addition of the PSCPM and ACBSS modules, there is a significant improvement in accuracy across all the indices. Comparing the first row and the second row, the addition of the PSCPM to the baseline model results in an improvement of 1.96% in mIoU and 3.5% in $F1$-score. Comparing the second row and the fourth row, the integration of the ACBSS strategy in the model results in an improvement of 3.9% in mIoU and 2.79% in $F1$-score.

TABLE I
Quantitative Results of the Ablation Study Related to the Proposed Techniques on the Second Dataset, Landsat-SCD Dataset, and Hi-UCD Mini Dataset With 20% Labeled Samples

| Proposed techniques | | SECOND | | Landsat-SCD | | Hi-UCD mini | |
|---|---|---|---|---|---|---|---|
| PSCPM | ACBSS | mIoU(%) | F1(%) | mIoU(%) | F1(%) | mIoU(%) | F1(%) |
| | | 21.47 | 31.36 | 58.41 | 71.23 | 28.50 | 38.49 |
| ✓ | | 22.18 | 32.20 | 64.20 | 76.82 | 30.46 | 41.99 |
| | ✓ | 22.26 | 31.90 | 63.31 | 76.26 | 30.57 | 40.05 |
| ✓ | ✓ | **24.17** | **34.86** | **66.11** | **78.47** | **34.36** | **44.78** |

Fig. 3 presents the visualization results for the ablation experiments on the SECOND dataset with 20% labeled data. The fourth and fifth columns represent the visualization under supervised conditions, while the sixth and seventh columns represent the visualization under semi-supervised conditions. Comparing the fourth and fifth columns of Fig. 3(a), it is clear that the PSCPM enhances the model's ability to perceive subtle changes, which is particularly evident in the

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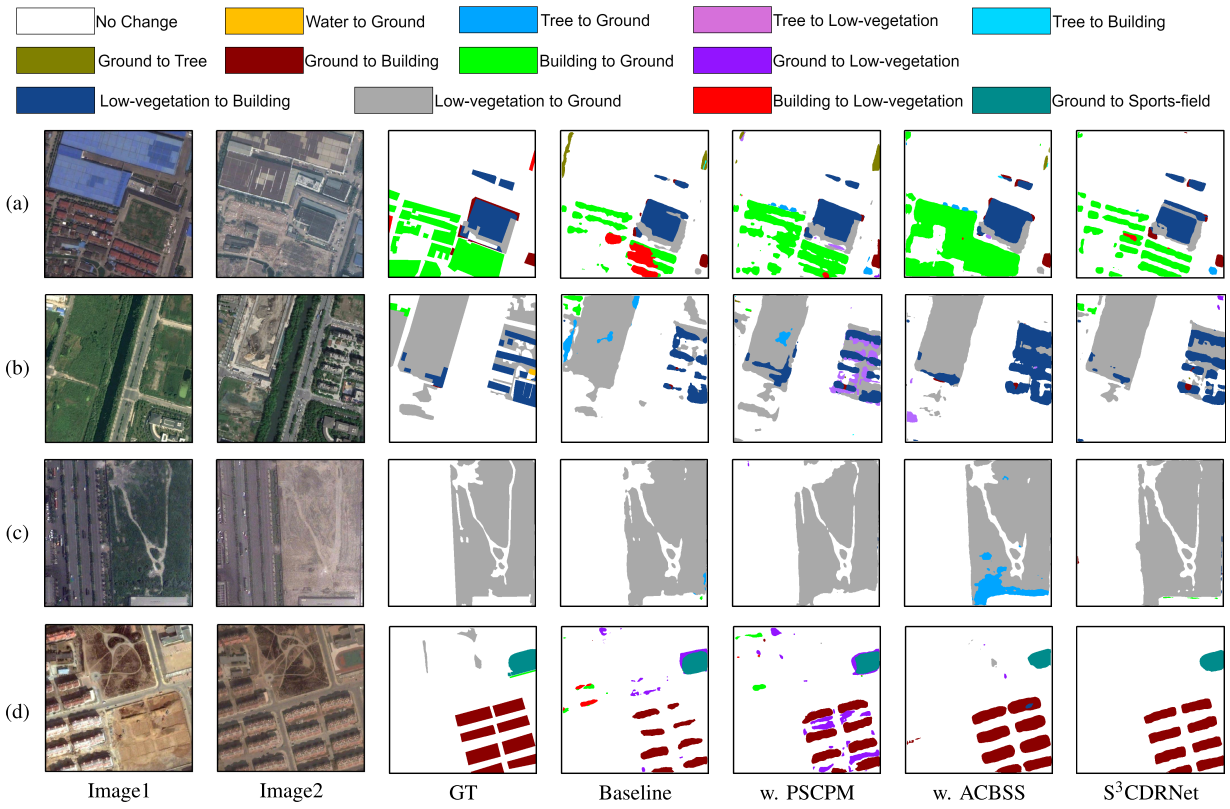



Fig. 3. Prediction results of the ablation experiments on the SECOND dataset with 20% labeled data. For each row, from left to right, are image T1, image T2, the ground truth, the prediction results of the baseline model, the prediction results of w. PSCPM (where w. PSCPM represents the integration of the PSCPM into the baseline model), the prediction results of w. ACBSS (where w. ACBSS represents the integration of the ACBSS semi-supervised learning strategy into the baseline model), and the prediction results of S$^3$CDRNet. (a)–(d) Visualizations of the samples selected from the SECOND dataset.

TABLE II

QUANTITATIVE RESULTS OF THE ABLATION STUDY RELATED TO THE IMAGE-LEVEL PERTURBATIONS ON THE SECOND DATASET AND LANDSAT-SCD DATASET WITH 20% LABELED SAMPLES

| Strong | Change class | SECOND | | Landsat-SCD | |
|---|---|---|---|---|---|
| augmention | confidence module | mIoU(%) | F1(%) | mIoU(%) | F1(%) |
| | | 20.95 | 29.97 | 53.39 | 66.70 |
| ✓ | | 23.41 | 33.49 | 64.45 | 77.16 |
| ✓ | ✓ | **24.17** | **34.86** | **66.11** | **78.47** |

clear boundaries and distinct intervals of the buildings. The comparison between the fourth and sixth columns shows that S$^3$CDRNet in the semi-supervised conditions achieves a more complete representation of the changed areas, although some detail is lost. Comparing the final two columns indicates that the proposed model balances the protection of details while providing a more complete representation of the changed areas in the semi-supervised conditions.

*2) Ablation Analysis of Image-Level Perturbation and the Change Class Confidence Module Within ACBSS Strategy:* In this analysis, the effectiveness of combining the strong image augmentation with the change class confidence module in the semi-supervised training strategy was analyzed. The experimental results are reported in Table II. The first row indicates the baseline semi-supervised method introduced in Section II, without using the strong image augmentation or the change class confidence module. Comparing the first row and second row, the addition of the strong image augmentation significantly improves the model generalization ability, resulting in an increase of 2.46% in mIoU and 3.52% in $F1$-score on the SECOND dataset, and an increase of 11.06% in mIoU and 10.46% in $F1$-score on the Landsat-SCD dataset. Comparing the second row and third row, the incorporation of the change class confidence module further filters the training samples by adjusting the sampling rate for strong data augmentation, resulting in an increase of 0.76% in mIoU and 1.37% in $F1$-score on the SECOND dataset, and an increase of 1.66% in mIoU and 1.31% in $F1$-score on the Landsat-SCD dataset. These results indicate that the image-level perturbations combined with the semi-supervised learning strategy can significantly improve the generalization ability of S$^3$CDRNet.

*3) Parameter Analysis:* The effects of different kernel sizes $M$ and $N$ in the PSCPM and pseudolabel threshold $\tau_p$ in the ACBSS strategy were analyzed on the SECOND dataset with 20% training samples. The experimental results are reported in Table III. When $M$ is set to 7, $N$ is set to 9, and the pseudolabel threshold $\tau_p$ is set to 0.7, S$^3$CDRNet achieves the highest accuracy, with the mIoU and $F1$-score reaching 24.17% and 34.86%, respectively.

## B. Comparison Results and Analysis

To validate the effectiveness of the proposed method, experiments were conducted under both supervised and

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TABLE III
QUANTITATIVE RESULTS OF S$^3$CDRNET ON THE SECOND DATASET WITH DIFFERENT $M$, $N$, AND $\tau_p$ VALUES

| $\tau_p$ values | $M$ and $N$ values | mIoU(%) | F1(%) |
|---|---|---|---|
| 0.3 | $M$ = 3, $N$ = 5 | 21.78 | 31.04 |
| | $M$ = 5, $N$ = 7 | 22.17 | 31.40 |
| | $M$ = 7, $N$ = 9 | 21.05 | 30.05 |
| | $M$ = 9, $N$ = 11 | 21.24 | 30.31 |
| 0.5 | $M$ = 3, $N$ = 5 | 22.55 | 32.55 |
| | $M$ = 5, $N$ = 7 | 22.13 | 31.51 |
| | $M$ = 7, $N$ = 9 | 21.47 | 30.82 |
| | $M$ = 9, $N$ = 11 | 21.30 | 30.22 |
| 0.7 | $M$ = 3, $N$ = 5 | 23.80 | 33.86 |
| | $M$ = 5, $N$ = 7 | 23.75 | 34.01 |
| | $M$ = 7, $N$ = 9 | **24.17** | **34.86** |
| | $M$ = 9, $N$ = 11 | 23.60 | 33.41 |
| 0.9 | $M$ = 3, $N$ = 5 | 21.67 | 31.08 |
| | $M$ = 5, $N$ = 7 | 22.40 | 31.99 |
| | $M$ = 7, $N$ = 9 | 21.13 | 30.39 |
| | $M$ = 9, $N$ = 11 | 21.12 | 30.33 |

semi-supervised conditions. To simulate conditions with insufficient labeled samples, we selected 5%, 10%, 15%, and 20% of the labeled samples in the training set. We compared the end-to-end networks of the fully convolutional network (FCN) [47], UNet++ [48], DeepLabv3+ [49], UNetFormer [50], the bitemporal image transformer (BIT) [4], and Pyramid-SCDFormer [36], along with the multitask methods of HRSCD-str.3 [19], HRSCD-str.4 [19], SSCDl [21], Bi-SRNet [21], SCanNet [22], and DEFO-MTLSCD [31]. Furthermore, different semi-supervised learning strategies—mean teacher [51] and SemiCD [7]—were also compared with the proposed semi-supervised strategy. We evaluated the accuracy of the SCD across these methods. For the multitask methods, the predictions were postprocessed to semantic change maps for accuracy evaluation.

*1) Results Analysis for the SECOND Dataset:* The accuracy performance of the various methods on the SECOND dataset is shown in Table IV. SCDRNet represents the proposed network without the semi-supervised strategy. Under supervised conditions, SCDRNet outperforms both the end-to-end and multitask networks in mIoU across 5%, 10%, 15%, and 20% labeled samples. Notably, with 5% labeled samples, the proposed method achieves the highest mIoU. Compared with SSCDl, although it acquires the highest $F1$-score at 26.51%, SCDRNet has a higher mIoU at 17.74%, which indicates a better visual result for the detailed changes. As the proportion of labeled samples increases, the accuracy also improves. With 10% labeled samples, the proposed method achieves the highest values in both mIoU and $F1$-score. For 15% and 20% labeled samples, the proposed method attains the best mIoU and the second-best $F1$-score, demonstrating a superior predictive ability. In the semi-supervised conditions, enhanced with the ACBSS strategy, the results show significant improvements across all the metrics. For the existing BIT algorithm, we tested its performance using the proposed semi-supervised strategy. With 5%, 10%, 15%, and 20% labeled samples, the accuracy improves by 2.43%, 5.29%, 2.64%, and 4.42% in mIoU, respectively. The semi-supervised strategy also enhances the performance of the proposed network—S$^3$CDRNet—which achieves the highest accuracy across all the metrics with 5%, 10%, and 15% labeled samples. Compared with the other semi-supervised strategies, S$^3$CDRNet achieves the highest mIoU and $F1$-score across all four proportions of labeled samples. When 5% labeled samples are used, S$^3$CDRNet outperforms SemiCD by 1.53% in mIoU and 1.68% in $F1$-score. As the number of labeled samples increases, the improvement in the accuracy of S$^3$CDRNet becomes more pronounced. As can be seen from the results for 10%, 15%, and 20% labeled samples, the improvement in the mIoU of S$^3$CDRNet over SemiCD reaches 2.75%, 3.54%, and 3.61%, respectively, and the corresponding $F1$-score reaches 3.4%, 4.24%, and 4.8%, respectively.

The visual results for 5% and 20% labeled samples on the SECOND dataset are shown in Figs. 4 and 5, respectively. As illustrated in Fig. 4, the proposed method demonstrates superior detail preservation under supervised conditions. For instance, in Fig. 4(b), the construction of new buildings is accurately detected. In Fig. 4(c), the proposed method maintains better road boundary integrity. Furthermore, in Fig. 4(d), the proposed method successfully detects the disappearance of small patches of low vegetation, showing its effectiveness in capturing such small region changes. Under semi-supervised conditions, the proposed approach shows significant visual improvements. As shown in Fig. 4(a), the proposed method effectively preserves detailed changes such as the demolition of small buildings. In Fig. 4(d), the proposed method accurately detects the construction of new buildings. With 20% labeled samples, the same samples as in Fig. 4 were selected. Due to the increase in labeled samples, the proposed method shows a significant improvement in visualization results. The visualization results of the semi-supervised method are superior to those of the other methods. The misdetections with 5% labeled samples are significantly reduced with 20% labeled samples. For example, in Fig. 5(d), the appearance of low vegetation is now accurately detected.

The confusion matrix for the SECOND dataset is shown in Fig. 6. Overall, compared with the other methods, the proposed method's predictions are more concentrated along the main diagonal. This indicates that the proposed method achieves a higher accuracy in predicting each change category.

*2) Results Analysis for the Landsat-SCD Dataset:* The accuracy performance of the various methods on the Landsat-SCD dataset is shown in Table V. Under supervised conditions, the proposed method achieves the best results in both mIoU and $F1$-score across 5%, 10%, 15%, and 20% labeled samples, indicating a superior performance in both detail and overall prediction. In semi-supervised conditions, the proposed method demonstrates a substantial improvement in prediction accuracy. The accuracy of the BIT model is significantly improved when combined with the ACBSS strategy. With 5% labeled samples, it achieves a 10.97% mIoU increase, compared with the supervised BIT model.

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TABLE IV

COMPARATIVE EXPERIMENTAL RESULTS OF DIFFERENT SCD METHODS WITH 5%, 10%, 15%, AND 20% LABELED SAMPLES ON THE SECOND DATASET, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

| Method | Training Strategies | Network Structure | 5% | | 10% | | 15% | | 20% | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | mIoU(%) | F1(%) | mIoU(%) | F1(%) | mIoU(%) | F1(%) | mIoU(%) | F1(%) |
| FCN [47] | Super.only | End-to-End | 10.70 | 15.71 | 12.23 | 17.95 | 13.02 | 19.39 | 14.26 | 21.26 |
| UNet++ [48] | Super.only | End-to-End | 10.98 | 15.80 | 11.27 | 16.05 | 13.24 | 19.24 | 13.58 | 19.72 |
| DeepLabv3+ [49] | Super.only | End-to-End | 9.88 | 14.21 | 11.92 | 17.61 | 12.90 | 18.86 | 14.35 | 21.37 |
| UNetFormer [50] | Super.only | End-to-End | 9.47 | 13.58 | 10.35 | 14.84 | 10.55 | 15.03 | 11.14 | 15.82 |
| BIT [4] | Super.only | End-to-End | 12.54 | 18.30 | 12.87 | 18.84 | 14.93 | 22.29 | 15.40 | 23.04 |
| Pyramid-SCDFormer [36] | Super.only | End-to-End | 11.74 | 16.79 | 12.93 | 19.03 | 13.62 | 19.91 | 14.54 | 21.35 |
| HRSCD-str.3 [19] | Super.only | Multi-task | 12.46 | 18.62 | 14.48 | 21.67 | 15.33 | 23.07 | 16.28 | 24.34 |
| HRSCD-str.4 [19] | Super.only | Multi-task | 13.97 | 20.82 | 16.35 | 24.38 | 20.79 | 31.12 | 18.96 | 28.02 |
| SSCDl [21] | Super.only | Multi-task | 17.48 | 26.51 | 18.48 | 27.59 | 19.89 | 29.55 | 19.73 | 29.56 |
| Bi-SRNet [21] | Super.only | Multi-task | 16.62 | 24.83 | 17.59 | 26.17 | 19.84 | 29.50 | 22.06 | **35.20** |
| SCanNet [22] | Super.only | Multi-task | 14.79 | 21.23 | 18.03 | 28.44 | 19.76 | 29.12 | 20.16 | 31.84 |
| DEFO-MTLSCD [31] | Super.only | Multi-task | 14.09 | 21.33 | 17.85 | 27.14 | 18.74 | 27.96 | 20.68 | 31.08 |
| SCDRNet | Super.only | End-to-End | 17.74 | 25.34 | 20.33 | 29.55 | 21.27 | 30.82 | 22.18 | 32.20 |
| BIT+ACBSS | Semi-super | End-to-End | 14.97 | 21.88 | 18.16 | 26.79 | 17.57 | 25.95 | 19.82 | 29.04 |
| SCDRNet+mean teacher [51] | Semi-super | End-to-End | 14.15 | 20.01 | 17.65 | 25.51 | 18.30 | 26.64 | 19.41 | 28.40 |
| SCDRNet+SemiCD [7] | Semi-super | End-to-End | 17.35 | 25.25 | 18.73 | 27.36 | 19.33 | 28.53 | 20.56 | 30.06 |
| $S^3$CDRNet | Semi-super | End-to-End | **18.88** | **26.93** | **21.48** | **30.76** | **22.87** | **32.77** | **24.17** | 34.86 |

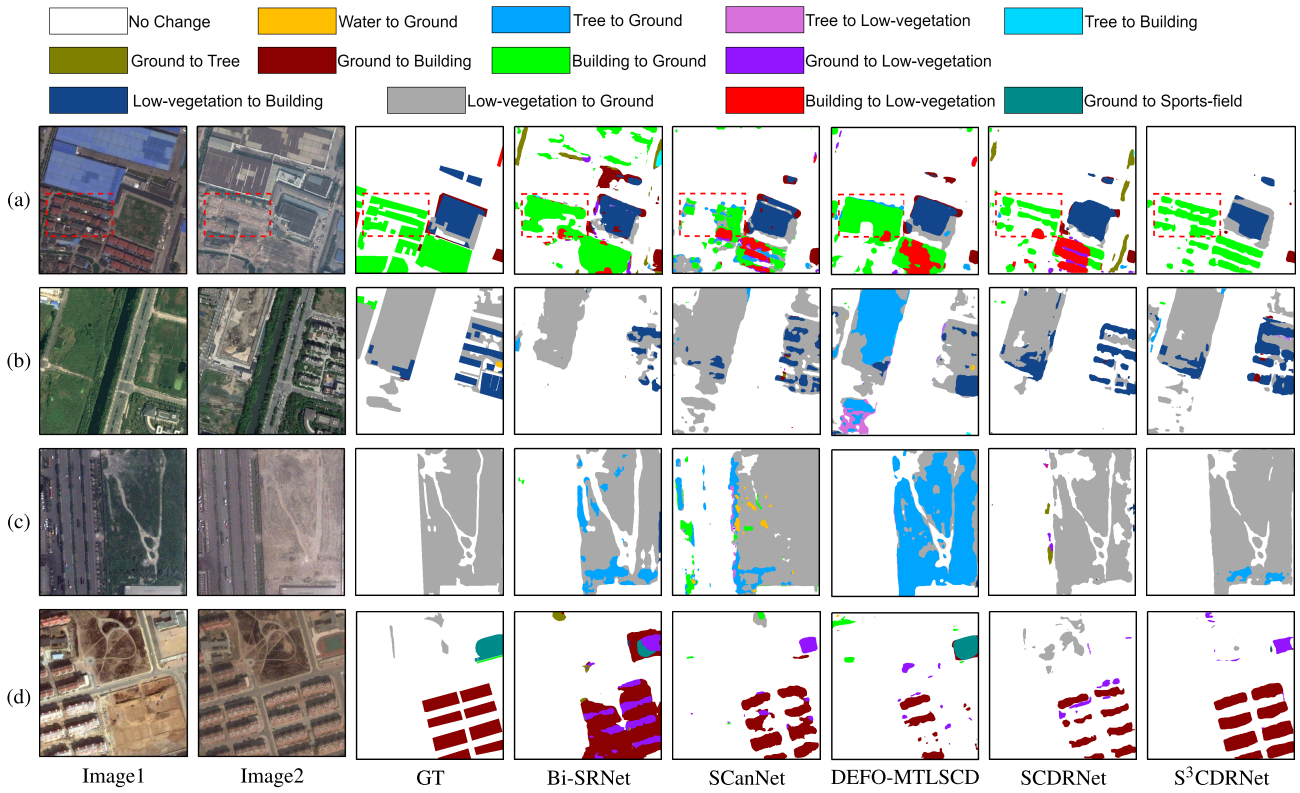



Fig. 4. Prediction results of different models on the SECOND dataset with 5% labeled data. For each row, from left to right, are image T1, image T2, the ground truth, the Bi-SRNet prediction results, the SCanNet prediction results, the DEFO-MTLSCD prediction results, the SCDRNet prediction results, and the $S^3$CDRNet prediction results. (a)–(d) Samples selected from the SECOND dataset.

As the number of labeled samples increases, the enhancement becomes more pronounced. With 20% labeled samples, the mIoU improvement reaches 19.61%, compared with the supervised BIT model. These results indicate the effectiveness of using semi-supervised learning for SCD tasks. Under semi-supervised conditions, $S^3$CDRNet achieves an mIoU

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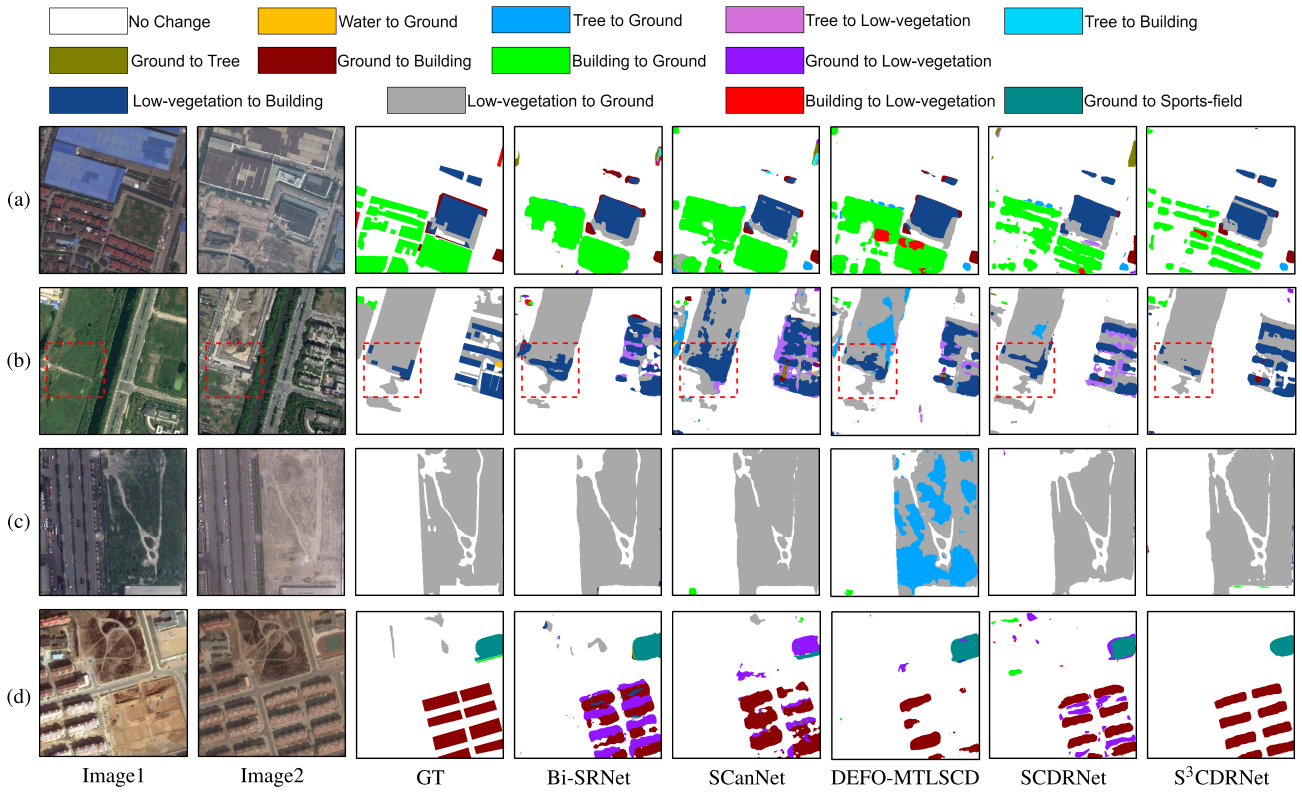



Fig. 5. Prediction results of the different models on the SECOND dataset with 20% labeled data. For each row, from left to right, are image T1, image T2, the ground truth, the Bi-SRNet prediction results, the SCanNet prediction results, the DEFO-MTLSCD prediction results, the SCDRNet prediction results, and the S3CDRNet prediction results. (a)–(d) Samples selected from the SECOND dataset.

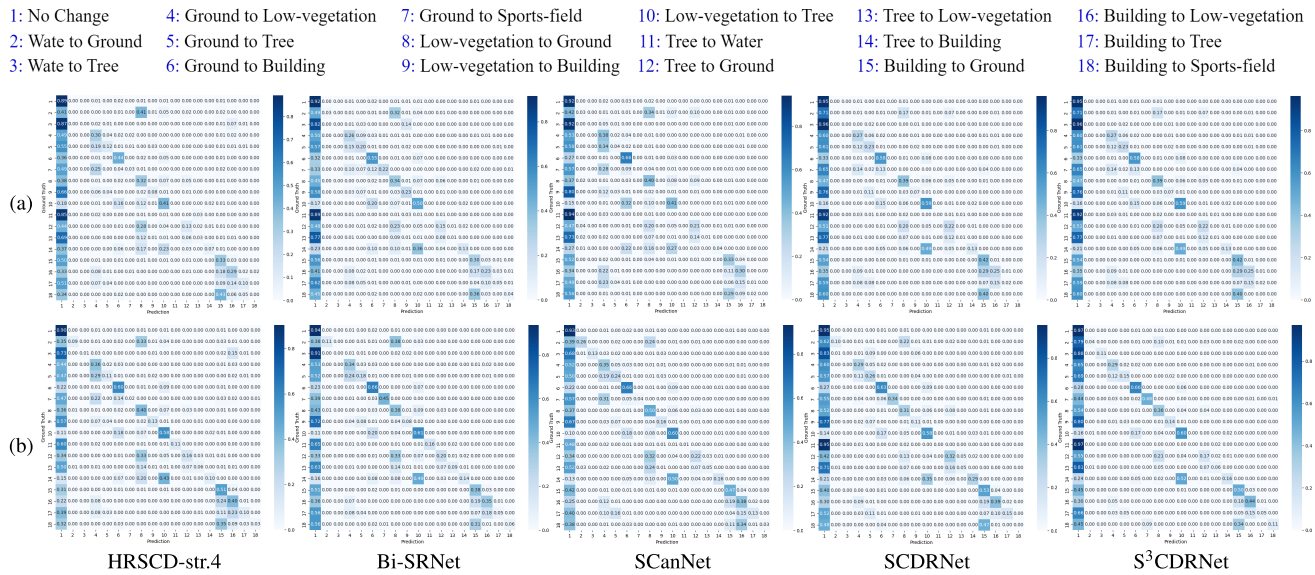



Fig. 6. Confusion matrix for each change class predicted by the different models with the SECOND dataset. (a) Experimental results under 5% labeled samples. (b) Experimental results under 20% labeled samples.

of 46.27%, 56.02%, 62.29%, and 66.11%, respectively, and an $F1$-score of 59.35%, 69.62%, 75.37%, and 78.47%, respectively. Compared with SemiCD, the proposed method shows improvements in mIoU of 3.86%, 4.37%, 6.01%, and 4.66%, respectively, and improvements in $F1$-score of 4.52%, 4.53%, 5.62%, and 3.9%, respectively.

The visual results for 5% and 20% labeled samples on the Landsat-SCD dataset are shown in Figs. 7 and 8, respectively. The supervised method demonstrates a clear advantage in detail preservation. For instance, in Fig. 7(a), the proposed method accurately detects the drying up of a river, capturing this detailed change comprehensively. In semi-supervised conditions, the ability to detect details is further enhanced. As shown in Fig. 7(c), even with just 5% labeled samples, the proposed method can detect the boundary details of farmland. With 20% labeled samples, this advantage in detail preservation becomes even more pronounced. For example, in Fig. 8(c), compared with Fig. 7(c) with 5% labeled samples,

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TABLE V

COMPARATIVE EXPERIMENTAL RESULTS OF DIFFERENT SCD METHODS WITH 5%, 10%, 15%, AND 20% LABELED SAMPLES ON THE LANDSAT-SCD DATASET, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

| Method | Training Strategies | Network Structure | 5% | | 10% | | 15% | | 20% | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | mIoU(%) | F1(%) | mIoU(%) | F1(%) | mIoU(%) | F1(%) | mIoU(%) | F1(%) |
| FCN [47] | Super.only | End-to-End | 33.61 | 43.68 | 35.84 | 47.00 | 38.12 | 49.60 | 40.50 | 52.46 |
| UNet++ [48] | Super.only | End-to-End | 30.73 | 39.51 | 34.72 | 44.07 | 37.65 | 47.43 | 39.75 | 50.00 |
| DeepLabv3+ [49] | Super.only | End-to-End | 32.41 | 41.80 | 35.40 | 45.10 | 37.21 | 47.57 | 39.10 | 50.07 |
| UNetFormer [50] | Super.only | End-to-End | 29.39 | 37.65 | 31.08 | 39.60 | 32.66 | 41.14 | 33.29 | 41.89 |
| BIT [4] | Super.only | End-to-End | 30.71 | 39.50 | 34.08 | 43.85 | 37.37 | 47.33 | 40.34 | 51.58 |
| Pyramid-SCDFormer [36] | Super.only | End-to-End | 29.26 | 38.02 | 33.12 | 41.92 | 35.21 | 45.08 | 39.50 | 50.61 |
| HRSCD-str.3 [19] | Super.only | Multi-task | 28.34 | 34.99 | 29.47 | 35.82 | 30.37 | 37.02 | 42.00 | 53.81 |
| HRSCD-str.4 [19] | Super.only | Multi-task | 28.68 | 36.15 | 34.84 | 44.84 | 37.87 | 49.13 | 45.95 | 58.74 |
| SSCDl [21] | Super.only | Multi-task | 35.52 | 47.20 | 43.59 | 56.84 | 46.78 | 60.32 | 53.06 | 67.07 |
| Bi-SRNet [21] | Super.only | Multi-task | 39.62 | 51.61 | 48.54 | 62.26 | 53.20 | 67.07 | 52.31 | 66.26 |
| SCanNet [22] | Super.only | Multi-task | 42.06 | 54.27 | 49.81 | 63.19 | 57.21 | 70.67 | 62.04 | 75.09 |
| DEFO-MTLSCD [31] | Super.only | Multi-task | 40.02 | 52.54 | 48.73 | 62.20 | 55.62 | 69.19 | 60.15 | 73.44 |
| SCDRNet | Super.only | End-to-End | 44.32 | 56.12 | 51.39 | 64.45 | 59.10 | 72.38 | 64.20 | 76.82 |
| BIT+ACBSS | Semi-super | End-to-End | 41.68 | 47.90 | 47.90 | 61.53 | 56.53 | 70.16 | 59.95 | 73.45 |
| SCDRNet+mean teacher [51] | Semi-super | End-to-End | 37.67 | 46.79 | 42.92 | 53.11 | 45.51 | 57.38 | 45.93 | 57.81 |
| SCDRNet+SemiCD [7] | Semi-super | End-to-End | 42.41 | 54.83 | 51.65 | 65.09 | 56.28 | 69.75 | 61.45 | 74.53 |
| $S^3$CDRNet | Semi-super | End-to-End | **46.27** | **59.35** | **56.02** | **69.62** | **62.29** | **75.37** | **66.11** | **78.47** |

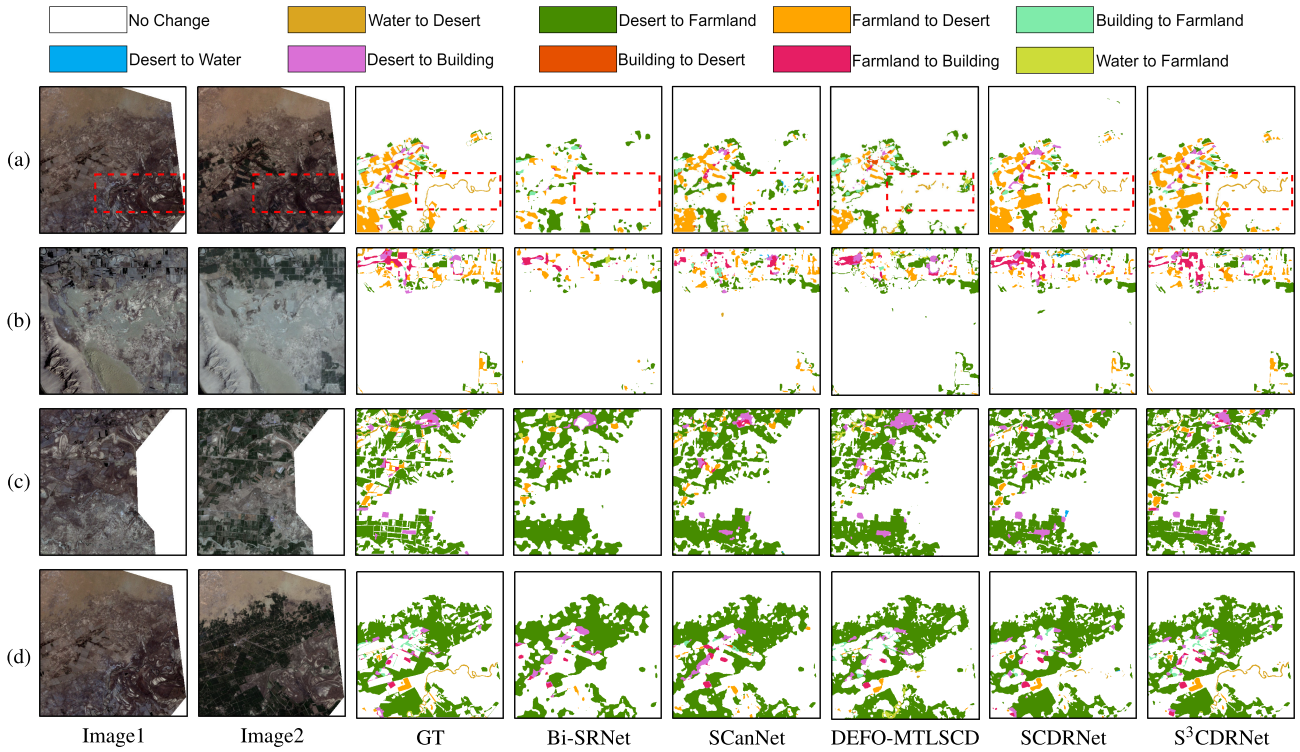



Fig. 7. Prediction results of different models on the Landsat-SCD dataset with 5% labeled data. For each row, from left to right, are image T1, image T2, the ground truth, the Bi-SRNet prediction results, the SCanNet prediction results, the DEFO-MTLSCD prediction results, the SCDRNet prediction results, and the $S^3$CDRNet prediction results. (a)–(d) Samples selected from the Landsat-SCD dataset.

the proposed model shows improved capability in preserving the boundary details of farmland.

The confusion matrix for the Landsat-SCD dataset is shown in Fig. 9. Compared with the SECOND dataset, there is a notable improvement in prediction accuracy. Compared with other methods, the proposed method's predictions are predominantly concentrated along the main diagonal, significantly reducing the misdetections for each category.

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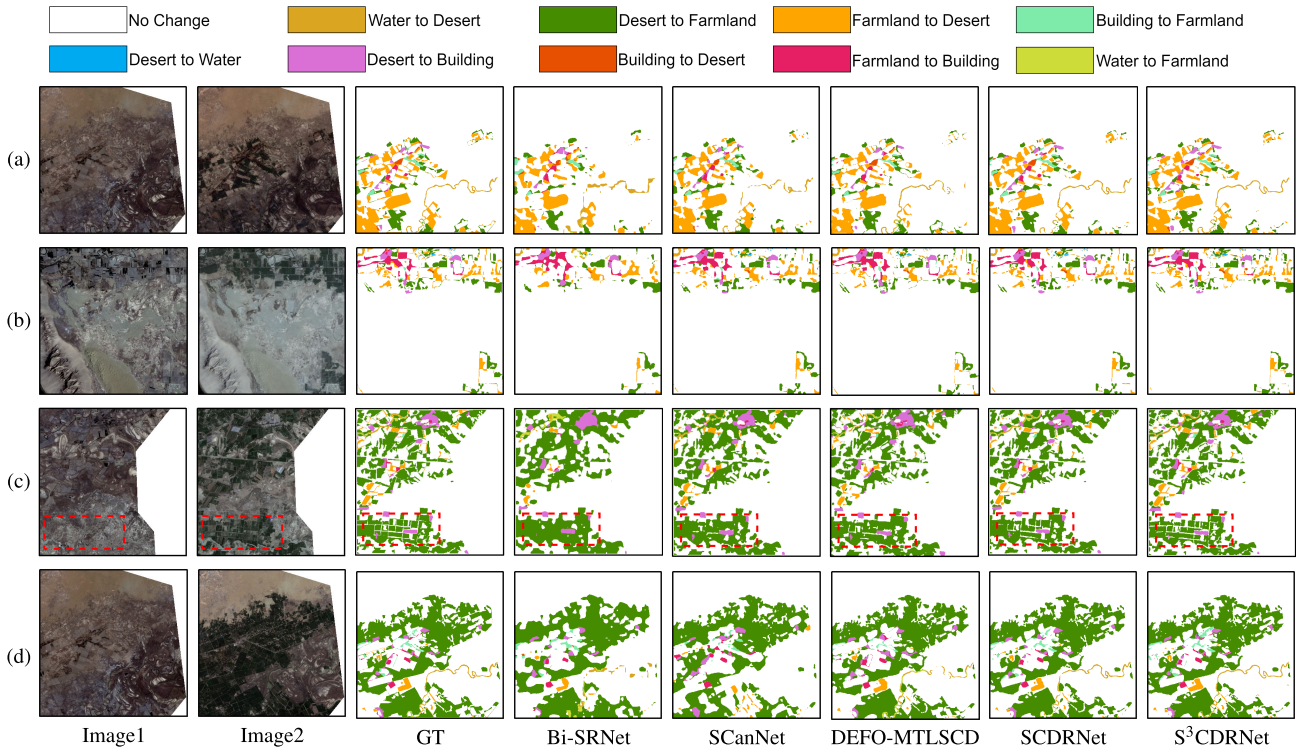



Fig. 8. Prediction results of different models on the Landsat-SCD dataset with 20% labeled data. For each row, from left to right, are image T1, image T2, the ground truth, the Bi-SRNet prediction results, the SCanNet prediction results, the DEFO-MTLSCD prediction results, the SCDRNet prediction results, and the S3CDRNet prediction results. (a)–(d) Samples selected from the Landsat-SCD dataset.

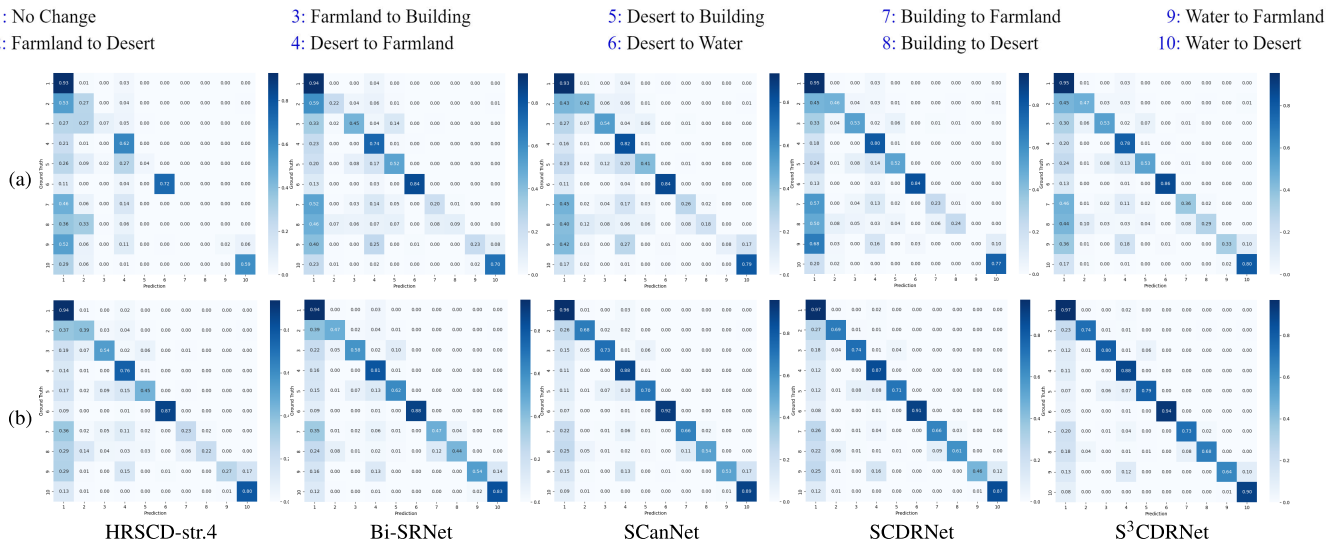



Fig. 9. Confusion matrix for each change class predicted by different models with the Landsat-SCD dataset. (a) Experimental results under 5% labeled samples. (b) Experimental results under 20% labeled samples.

This demonstrates that the proposed method significantly enhances the prediction accuracy for the tail classes, compared with other approaches, effectively mitigating the issue of class imbalance.

*3) Results Analysis for the Hi-UCD Mini Dataset:* The accuracy performance of different methods on the Hi-UCD mini dataset is shown in Table VI. Under supervised conditions, SCDRNet performs less well than the multitask model when only 5% labeled samples are used, indicating that the multitask model shows a better generalization ability when dealing with a small proportion of labeled samples on this dataset. It is worth noting that when using 10%, 15%, and 20% labeled samples, the proposed SCDRNet outperforms the comparison methods in both mIoU and $F1$-score, demonstrating its superiority in change detail preservation and overall performance. Under semi-supervised conditions, the introduction of the ACBSS semi-supervised strategy significantly improves all the metrics of S$^3$CDRNet. When 5% labeled samples are used, although DEFO-MTLSCD achieves the highest $F1$-score at 31.23%, S$^3$CDRNet has a higher mIoU at 22.97%, providing better visual effects for change details. Compared with other semi-supervised learning methods, the proposed semi-supervised learning strategy still maintains a

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TABLE VI

COMPARATIVE EXPERIMENTAL RESULTS OF DIFFERENT SCD METHODS WITH 5%, 10%, 15%, AND 20% LABELED SAMPLES ON THE HI-UCD MINI DATASET, WITH THE HIGHEST SCORES HIGHLIGHTED IN BOLD

| Method | Training Strategies | Network Structure | 5% | | 10% | | 15% | | 20% | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | mIoU(%) | F1(%) | mIoU(%) | F1(%) | mIoU(%) | F1(%) | mIoU(%) | F1(%) |
| FCN [47] | Super.only | End-to-End | 15.06 | 19.03 | 17.97 | 23.73 | 18.89 | 25.10 | 20.77 | 28.17 |
| UNet++ [48] | Super.only | End-to-End | 14.13 | 16.53 | 15.46 | 19.19 | 17.81 | 22.40 | 18.35 | 23.79 |
| DeepLabv3+ [49] | Super.only | End-to-End | 14.92 | 19.78 | 16.07 | 20.76 | 17.67 | 24.01 | 20.18 | 26.97 |
| UNetFormer [50] | Super.only | End-to-End | 12.16 | 13.49 | 13.27 | 15.37 | 17.25 | 22.10 | 17.50 | 23.74 |
| BIT [4] | Super.only | End-to-End | 15.63 | 19.52 | 18.79 | 24.21 | 20.74 | 27.76 | 22.31 | 29.36 |
| Pyramid-SCDFormer [36] | Super.only | End-to-End | 14.15 | 17.14 | 15.77 | 18.90 | 19.49 | 25.97 | 22.50 | 30.62 |
| HRSCD-str.3 [19] | Super.only | Multi-task | 18.63 | 26.61 | 19.91 | 27.70 | 20.19 | 27.93 | 21.47 | 30.55 |
| HRSCD-str.4 [19] | Super.only | Multi-task | 21.76 | 30.59 | 23.39 | 32.75 | 23.68 | 33.23 | 25.34 | 35.71 |
| SSCDl [21] | Super.only | Multi-task | 20.08 | 27.62 | 21.35 | 29.50 | 25.46 | 36.17 | 26.58 | 36.58 |
| Bi-SRNet [21] | Super.only | Multi-task | 18.42 | 26.03 | 23.71 | 32.96 | 23.80 | 33.02 | 28.31 | 38.71 |
| SCanNet [22] | Super.only | Multi-task | 18.97 | 25.64 | 19.48 | 26.52 | 23.99 | 33.38 | 25.87 | 36.04 |
| DEFO-MTLSCD [31] | Super.only | Multi-task | 22.42 | **31.23** | 22.66 | 31.75 | 22.82 | 32.33 | 25.27 | 35.33 |
| SCDRNet | Super.only | End-to-End | 19.13 | 25.76 | 24.58 | 33.27 | 27.43 | 36.59 | 30.46 | 41.99 |
| BIT+ACBSS | Semi-super | End-to-End | 19.12 | 25.63 | 22.22 | 30.01 | 26.39 | 35.97 | 29.22 | 39.23 |
| SCDRNet+mean teacher [51] | Semi-super | End-to-End | 12.44 | 13.96 | 17.52 | 21.65 | 22.73 | 28.92 | 26.43 | 34.39 |
| SCDRNet+SemiCD [7] | Semi-super | End-to-End | 15.06 | 18.91 | 22.64 | 30.61 | 26.56 | 35.86 | 28.28 | 38.07 |
| $S^3$CDRNet | Semi-super | End-to-End | **22.97** | 31.03 | **25.84** | **34.70** | **31.73** | **41.97** | **34.36** | **44.78** |

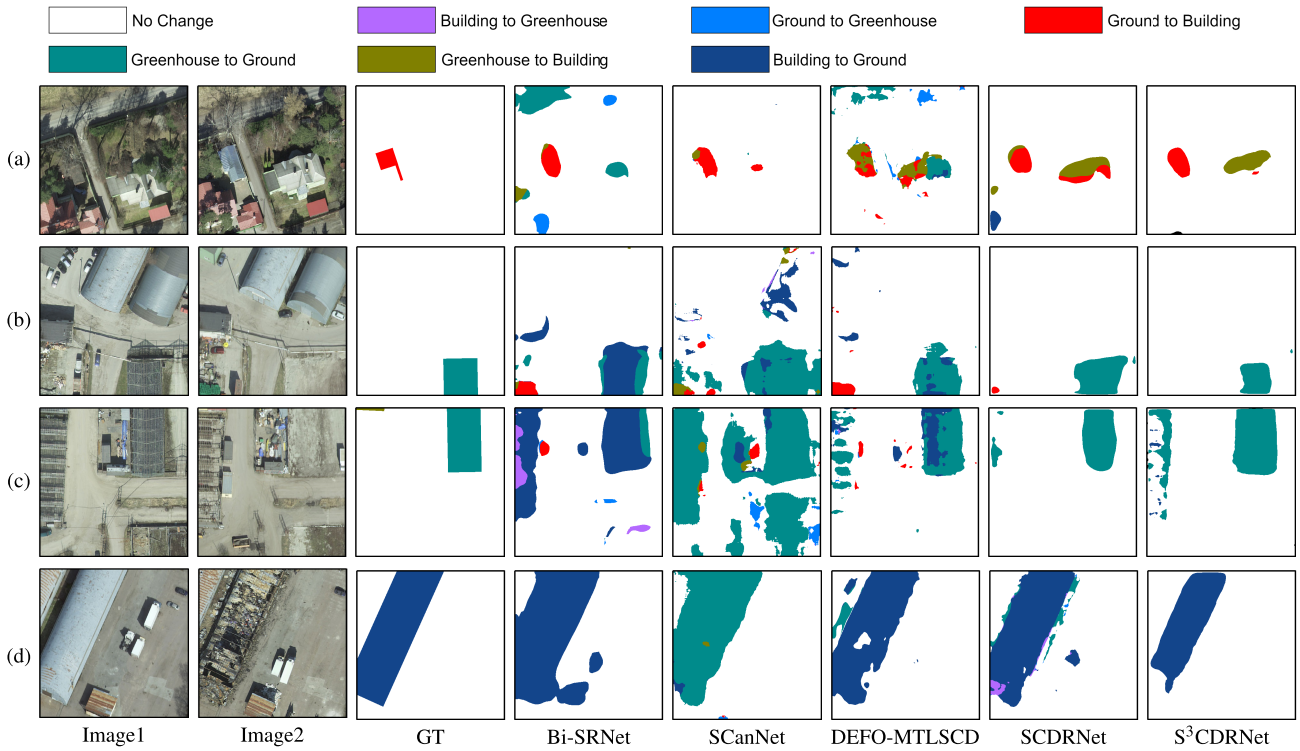



Fig. 10. Prediction results of different models on the Hi-UCD mini dataset with 5% labeled data. For each row, from left to right, are image T1, image T2, the ground truth, the Bi-SRNet prediction results, the SCanNet prediction results, the DEFO-MTLSCD prediction results, the SCDRNet prediction results, and the $S^3$CDRNet prediction results. (a)–(d) Samples selected from the Hi-UCD mini dataset.

good performance. When 10%, 15%, and 20% labeled samples are used, the proposed semi-supervised learning strategy achieves the best accuracy in all the indices.

The visual results of the Hi-UCD mini dataset with 5% and 20% labeled samples are shown in Figs. 10 and 11, respectively. The proposed method has advantages in detail preservation under supervised conditions. For example, in Fig. 10(b) and (c), the proposed method can accurately detect "demolition of the greenhouse." Comparing Figs. 10(c) and 11(c), as the number of labeled samples

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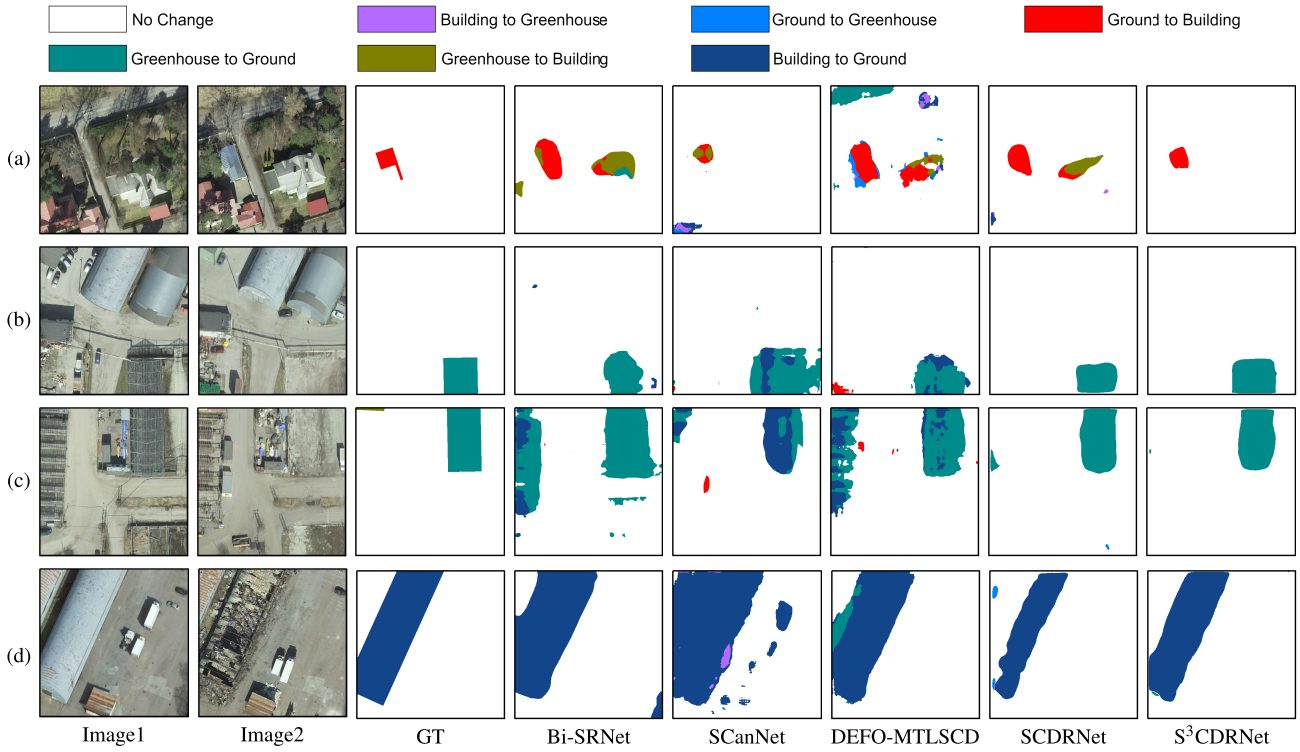



Fig. 11. Prediction results of different models on the Hi-UCD mini dataset with 20% labeled data. For each row, from left to right, are image T1, image T2, the ground truth, the Bi-SRNet prediction results, the SCanNet prediction results, the DEFO-MTLSCD prediction results, the SCDRNet prediction results, and the S3CDRNet prediction results. (a)–(d) Samples selected from the Hi-UCD mini dataset.

increases, the model's ability to detect details is further enhanced. Under semi-supervised conditions, as shown in Figs. 10(d) and 11(d), when 5% labeled samples are used, S$^3$CDRNet detects more complete boundary information of demolished buildings. When 20% labeled samples are used, S$^3$CDRNet provides a more complete presentation of changed areas.

## V. Conclusion

In this article, to address the inherent issues of spatial size imbalance and change category imbalance in SCD tasks, we have proposed S$^3$CDRNet for SCD. First, we designed a Siamese end-to-end SCD network (SCDRNet) that combines the proposed PSCPM module based on large kernel convolution to enhance the extraction of semantic change information in small change regions. Second, we developed the ACBSS strategy based on the mean teacher framework to consider the long-tail problem in SCD. The change class confidence module is used to record the learning status of each class and to control the data augmentation sampling rate to provide more information for the tail classes. In the experiments with small proportions of labeled samples on three public SCD datasets, the proposed S$^3$CDRNet outperformed the existing state-of-the-art methods in preserving change details and mitigating the issue of class imbalance. During the semi-supervised training phase of S$^3$CDRNet, a fixed threshold is used for pseudolabel generation. A higher threshold ensures the quality of pseudolabels, while a lower threshold increases the risk of noise and affects the model convergence speed. An adaptive thresholding method for the pseudolabels will be explored in our future work.

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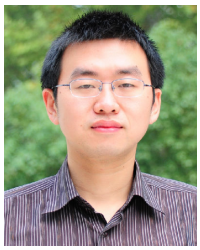
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